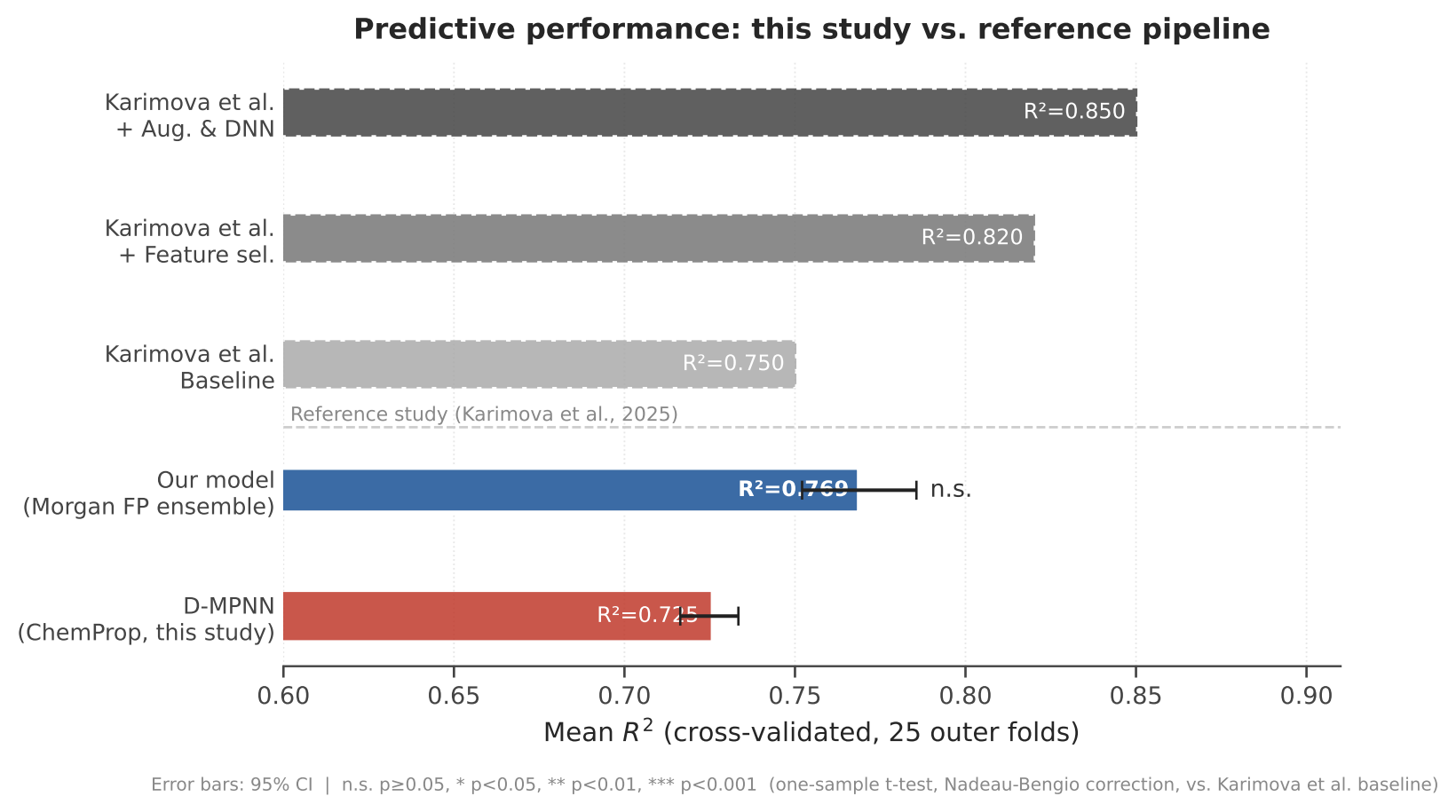
Predictive performance: this study vs. reference pipeline
Karimova et al.
+ Aug. & DNN
R²=0.850
Karimova et al.
+ Feature sel.
R²=0.820
Karimova et al.
Baseline
R²=0.750
Reference study (Karimova et al., 2025)
Our model
(Morgan FP ensemble)
R²=0.769
n.s.
D-MPNN
(ChemProp, this study)
R²=0.715
0.60
0.65
0.70
0.75
0.80
0.85
0.90
Mean R^2 (cross-validated, 25 outer folds)
Error bars: 95% CI | n.s. p≥0.05, * p<0.05, ** p<0.01, *** p<0.001 (one-sample t-test, Nadeau-Bengio correction, vs. Karimova et al. baseline)